

Fleet Analytics Technical Report

1. Objective

Analyze one week of VexarDrive fleet data and build two dashboards: a Driver Behaviour Dashboard to identify safe and risky driving patterns, and a Vehicle Health Status Dashboard to identify vehicles that may require maintenance based on trip and sensor data.

2. Dataset Overview

The dataset covers 30 drivers, 30 vehicles, 450 trips, and 12,987 telemetry records collected during one week. Excel was used for calculations and Power BI was used for dashboard development.

Table	Rows
Drivers	30
Vehicles	30
Trips	450
Telemetry Records	12,987

3. Driver Behaviour Dashboard

The Driver Behaviour Dashboard evaluates the riding behaviour of all 30 drivers during one week by combining trip data with mobile sensor readings. Instead of relying on a single metric, the dashboard combines speeding, braking, acceleration, and turning behaviour into one overall safety score.

Dashboard KPIs with justification

Dashboard Value	Why this value appears
450 Total Trips	The Trips table contains 450 completed trips, meaning every trip completed during the week is included in the analysis.
24.29 km/h Average Trip Speed	This is the average of the Avg_Speed_kmph recorded for all 450 trips. It represents normal fleet operating speed rather than a single maximum speed.
402 Total Overspeed Events	Every telemetry record where Speed_kmph > 45 was counted as an overspeed event. Across the week, this condition occurred 402 times.
63.51 Average Driver Safety Score	After calculating the Safety Score for each of the 30 drivers, the average score across the fleet is 63.51, indicating that the fleet generally falls in the safer range while still containing drivers who require attention.

How the Driver Safety Score was calculated

Four driving events were measured from the telemetry data.

Event	Rule
Overspeed	Speed_kmph > 45
Harsh Braking	Accel_X_g < -0.4
Harsh Acceleration	Accel_X_g > 0.4
Sharp Turn	ABS(Gyro_Z_dps) > 30

Since these metrics have different scales, each one was normalized using the highest observed value.

The final Safety Score was calculated as:

$$100 - (0.35 \times \text{Overspeed} + 0.25 \times \text{Braking} + 0.20 \times \text{Acceleration} + 0.20 \times \text{Turning})$$

Why these weights were chosen

- Overspeeding received the highest weight (35%) because repeated speeding has the greatest safety impact.
- Harsh braking (25%) indicates sudden stopping behaviour.
- Harsh acceleration and sharp turns (20% each) capture aggressive riding patterns.

Higher scores indicate safer driving because unsafe events reduce the final score.

Driver insights

- The fleet completed 450 trips during the week with an average trip speed of 24.29 km/h, indicating a relatively controlled operating speed overall.
- 402 overspeed events were recorded, making speeding the most significant behavioural risk observed in the dataset.
- The average Driver Safety Score is 63.51, showing that overall fleet behaviour is closer to the safer range while still containing drivers with consistently risky patterns.
- The Driver Safety Category Distribution shows:
 - 43.33% Moderate
 - 30.00% Safe

- 26.67% Risky
- Drivers with the highest overspeed counts consistently appear among the lowest Safety Scores, showing that overspeeding had the greatest influence on the final ranking.
- The Driver Safety Performance Summary allows every driver's score to be traced directly back to overspeed, braking, acceleration, and turning events, making the scoring transparent.

4. Vehicle Health Status Dashboard

The Vehicle Health Dashboard estimates the relative condition of each vehicle by combining sensor-derived vibration, rotational movement, vehicle age, and odometer readings.

Dashboard KPIs with justification

Dashboard Value	Why this value appears
23.00 Average Health Score	Average Health Score across all 30 vehicles after combining four normalized health indicators.
30 Total Vehicles	Count of unique Vehicle_ID values, confirming that every vehicle in the dataset is included.
30.64 Average Vibration	Average vibration magnitude calculated from three-axis accelerometer readings.
191.05 Average Gyro Instability	Average combined rotational movement calculated from all three gyroscope axes.

How the Health Score was calculated

Two additional sensor metrics were created.

Vibration Magnitude

$$= \text{SQRT} ([\text{Accel_X_g}]^2 + [\text{Accel_Y_g}]^2 + [\text{Accel_Z_g}]^2)$$

This combines movement across all three accelerometer axes into one vibration indicator.

Gyro Instability

$$|\text{GyroX}| + |\text{GyroY}| + |\text{GyroZ}|$$

This measures overall rotational movement.

Vehicle Age

$$2026 - \text{Manufacture Year}$$

Health Score

All four metrics were normalized before calculating the final score.

$$100 - (0.40 \times \text{Vibration} + 0.25 \times \text{Gyro} + 0.20 \times \text{Age} + 0.15 \times \text{Odometer})$$

Why these weights?

- Vibration received the highest weight because abnormal vibration is often associated with mechanical wear.
- Gyroscope instability captures unusual vehicle movement.
- Vehicle age and odometer provide supporting maintenance context.

Vehicle insights

- The fleet's average Health Score is 23.00, meaning several vehicles rank noticeably lower than others and should receive higher maintenance priority.
- Vehicles V02, V12, and V10 appear at the bottom of the Health Score ranking, making them the strongest candidates for inspection.
- The Vehicle Age vs Health Score scatter plot shows that older vehicles generally appear lower on the Health Score scale, suggesting that age contributes to declining health.
- The maintenance distribution shows:
 - 53.33% Monitor
 - 30.00% Needs Maintenance
 - 16.67% Healthy
- Comparing Health Score with odometer readings helps distinguish vehicles that combine higher mileage with lower health scores.

5. Assumptions

- Since the dataset did not specify an official fleet speed limit, 45 km/h was used as the overspeed threshold.
- ± 0.4 g was used to identify harsh acceleration and braking events.
- 30 dps was used as the threshold for sharp turning.
- Vibration and gyroscope values were treated as indicators of possible mechanical wear rather than confirmed faults.
- Vehicle health scores are intended to rank vehicles relative to each other within this dataset, not replace a physical inspection.

6. Additional Uses for the Dataset

Use Case	Business Value
Driver coaching	Identify riders who need additional safety training.
Predictive maintenance	Schedule servicing before breakdowns occur.
Route optimization	Compare driving behaviour across different delivery routes.
Insurance scoring	Support usage-based insurance pricing.
Fuel efficiency analysis	Identify driving habits that increase fuel consumption.
Fleet utilization	Understand whether vehicles are being used evenly.
Driver incentive programs	Reward consistently safe riders.
Accident risk monitoring	Detect risky behaviour before incidents occur.

7. Conclusion

The dashboards combine trip-level data with mobile sensor readings to provide practical insights into driver behaviour and vehicle condition. Together, they provide a useful foundation for improving safety, maintenance planning, and overall fleet operations.